GROUP 11 (high school biology)

Question UUID:

6015d7ef-c598-4e4f-9cfe-6c678e5a1a21

Question:

The sight organs of crustaceans and insects contain ommatidia, which make up the individual visual units of the

Options:

Choice_A: eyespot

Choice_B: simple eye

Choice_C: compound eye

Choice_D: binocular eye

Model:

meta-llama/Llama-3.1-8B-Instruct

Generated Reasoning Trace:

Step 1: Let's start by examining Choice_A, "eyespot." Eyespots are typically simple light-sensitive organs found in various organisms, often used for detecting polarized light or brightness. They do not possess the complex structure necessary to form a large visual field or to distinguish between different directions of light.

Step 2: Moving on to Choice_B, "simple eye." Simple eyes, such as the ones found in flatworms and some mollusks, usually consist of a single light-sensitive structure, like a

photoreceptor or a lens. They lack the complexity and organization to form a large number of visual units.

Step 3: Next, we have Choice_C, "compound eye." Compound eyes, found in arthropods such as insects and crustaceans, are composed of many small individual visual units, called ommatidia. Each ommatidium is made up of photoreceptors, and the large number of ommatidia in the compound eye allows for a wide field of view and enhanced detection of movement.

Step 4: Finally, let's examine Choice_D, "binocular eye." Binocular eyes are typically found in vertebrates, like humans and some fish. They provide a single, unified visual field, achieved by having two eyes positioned close together and overlapping their fields of view.

Step 5: Now, considering the composition of the sight organs in crustaceans and insects, it's clear that they contain numerous small, individual visual units. This specific arrangement is characteristic of compound eyes. Therefore, the correct answer is the choice that describes these complex structures composed of multiple visual units.

Step 6: Upon analysis, it's evident that only Choice_C, "compound eye," accurately describes the structure of the sight organs in crustaceans and insects, which contain ommatidia making up the individual visual units of the eyes.

Solution: Choice_C

Correct Answer: C

Commitment extracted
Logical Principle: Compound eyes in arthropods are defined by their ommatidia.
Explicit: "Eyespots are typically simple light-sensitive organs found in various organisms"; "Simple eyes usually consist of a single light-sensitive structure"; "Compound eyes, found in arthropods such as insects and crustaceans, are composed of many small individual visual units, called ommatidia"; "Binocular eyes are typically found in vertebrates"; "Binocular eyes are achieved by having two eyes positioned close together".
Implicit: Choice A and Choice B are eliminated because they describe singular or non-

modular structures lacking repeating multicellular arrays; Choice D is eliminated because binocular vision relies on the spatial orientation and overlapping fields of view of camera-type eyes rather than an assembly of microscopic optical facets; the deductive analysis confirms that any arthropod visual system made of grouped ommatidial facets must be a compound eye.

Question UUID:

e6a2f560-9a22-4613-90db-958234a19522

Question:

In fungi, cell walls are reinforced by which of the following?

Options:

Choice_A: Chitin

Choice_B: Silica

Choice_C: Lignin

Choice_D: Collagen fibers

Model:

Qwen2.5-7B-Instruct

Generated Reasoning Trace:

Step 1: Chitin is a major component of fungal cell walls. It provides structural support and helps maintain the shape of the cells. Therefore, it is a strong candidate.

Step 2: Silica is an inorganic substance often found in plant cell walls but not typically in fungal cell walls.

Step 3: Lignin is primarily found in the secondary cell walls of vascular plants, particularly in the cell walls of xylem tissues, and is not a characteristic component of fungal cell walls.

Step 4: Collagen fibers are found in animal tissues and are not present in fungi.

Step 5: Based on the above analysis, chitin is the correct component that reinforces fungal cell walls.

Solution: Choice_A

Correct Answer: A

Commitment extracted
Logical Principle: Chitin is found in fungi when the others are not.
Explicit: "Chitin is a major component of fungal cell walls"; "It provides structural support and helps maintain the shape of the cells"; "Silica is an inorganic substance often found in plant cell walls but not typically in fungal cell walls"; "Lignin is primarily found in the secondary cell walls of vascular plants"; "Collagen fibers are found in animal tissues and are not present in fungi".
Implicit: Choices B and C are eliminated because silica and lignin are exclusive to plant lineages and specific protists; Choice D is eliminated because collagen is found in animal tissie; the text establishes that chitin microfibrils provide the high tensile strength required for fungal morphological integrity.

Question UUID:

15454c85-3997-4724-8f65-40e04a51cd83

Question:

Mammals are homeostatic for all of the following EXCEPT,

Options:

Choice_A: body temperature

Choice_B: blood glucose concentration

Choice_C: blood pH

Choice_D: metabolic rate

Model:

Qwen3.6-27B

Generated Reasoning Trace:

Step 1: Evaluate Choice A, body temperature. Mammals are endotherms that actively regulate their internal body temperature through mechanisms like sweating, shivering, and vasodilation/vasoconstriction to maintain it within a narrow, optimal range, making it a classic example of homeostasis.

Step 2: Evaluate Choice B, blood glucose concentration. Mammals tightly regulate blood sugar levels using hormones such as insulin and glucagon to ensure cells receive adequate energy while preventing toxic highs or dangerous lows, which is a fundamental homeostatic process.

Step 3: Evaluate Choice C, blood pH. Mammals maintain blood pH within a very narrow range (approximately 7.35 to 7.45) through chemical buffers, respiratory adjustments, and renal excretion of acids or bases, demonstrating strict homeostatic control.

Step 4: Evaluate Choice D, metabolic rate. Unlike the previous parameters, metabolic rate in mammals is not maintained at a constant set point; it fluctuates significantly based on

factors such as physical activity, digestion, growth, reproductive state, and environmental demands, meaning it is not strictly homeostatic.

Step 5: Compare all choices to determine the exception. Since body temperature, blood glucose concentration, and blood pH are all tightly regulated to remain constant, while metabolic rate naturally varies to meet the organism's changing energy demands, metabolic rate is the correct exception.

Solution: Choice_D

Correct Answer: D

Commitment extracted

Logical Principle: Homeostatic parameters operate around tightly controlled, invariant physiological set points, whereas metabolic rate acts as a dynamic effect.

Explicit: "Mammals are endotherms that actively regulate their internal body temperature, making it a classic example of homeostasis"; "Mammals tightly regulate blood sugar levels using hormones such as insulin and glucagon, which is a fundamental homeostatic process"; "Mammals maintain blood pH within a very narrow range, demonstrating strict homeostatic control"; "Metabolic rate in mammals is not maintained at a constant set point; it fluctuates significantly".

Implicit: Choices A, B, and C are eliminated because core body temperature, blood glucose levels, and blood pH possess strict physiological boundaries regulated by distinct homeostatic baselines; Choice D represents the exception because metabolism is an adjustable process rather than a parameter constrained to a fixed internal target value.

GROUP 13 (college medicine)

Multiple Choice Questions Review

Q1 - Llama 3.1 8B

Question

After being severely beaten and sustaining a gunshot wound to the abdomen, a 42-year-old woman undergoes resection of a perforated small bowel. During the operation, plastic reconstruction of facial fractures, and open reduction and internal fixation of the left femur are also done. Thirty-six hours postoperatively, she is awake but not completely alert. She is receiving intravenous morphine via a patient-controlled pump. She says that she needs the morphine to treat her pain, but she is worried that she is becoming addicted. She has no history of substance use disorder. She drinks one to two glasses of wine weekly.

Which of the following initial actions by the physician is most appropriate?

Options

A. Reassure the patient that her chance of becoming addicted to narcotics is minuscule
B. Maintain the morphine, but periodically administer intravenous naloxone
C. Switch the patient to oral acetaminophen as soon as she can take medication orally
D. Switch the patient to intramuscular lorazepam

Correct Answer

A. Reassure the patient that her chance of becoming addicted to narcotics is minuscule

Reasoning

- The patient is experiencing severe postoperative pain after multiple major surgeries.
- Short-term opioid use for acute pain management in patients without a history of substance use disorder carries a very low risk of addiction.
- Reassurance and appropriate pain control are the best initial management steps.

Why the Other Options Are Incorrect

- **B:** Naloxone is used to reverse opioid overdose, not prevent addiction.
- **C:** Acetaminophen alone is insufficient for severe postoperative pain.
- **D:** Lorazepam is a benzodiazepine and is not appropriate for pain management.

Commitment extracted
Logical Principle: In acute post-operative pain management for major trauma, the risk of developing an opioid use disorder is exceptionally low in patients without a history of substance abuse, making immediate patient education and therapeutic reassurance the standard initial intervention.

Explicit: “The patient is experiencing severe postoperative pain after multiple major surgeries”; “Short-term opioid use for acute pain management in patients without a history of substance use disorder carries a very low risk of addiction”; “Reassurance and appropriate pain control are the best initial management steps”; “Naloxone is used to reverse opioid overdose, not prevent addiction”; “Acetaminophen alone is insufficient for severe postoperative pain”; “Lorazepam is a benzodiazepine and is not appropriate for pain management”.
Implicit: Choice B is eliminated because administering an opioid antagonist during therapeutic dosing precipitates withdrawal and destroys necessary analgesia; Choice C is eliminated because step-down monotherapy fails to match the severe nociceptive load of multi-site surgical trauma; Choice D is eliminated because anxiolytics lack direct analgesic efficacy; the diagnostic history validates option A as the ethically and clinically sound mechanism to preserve patient trust while maintaining adequate analgesia.

Q2 - Qwen 2.5 7B

Question

Selective Androgen Receptor Modulators (SARMs) are:

Options

A. Steroid drugs that act on androgen receptors mimicking the effects of natural steroid hormones
B. Steroid drugs that act on androgen receptors antagonising the effects of natural steroid hormones
C. Non-steroid drugs that act on androgen receptors mimicking the effects of natural steroid hormones
D. Non-steroid drugs that act on androgen receptors inhibiting the effects of natural steroid hormones

Correct Answer

C. Non-steroid drugs that act on androgen receptors mimicking the effects of natural steroid hormones

Reasoning

- SARMs selectively activate androgen receptors in specific tissues.
- They are designed to produce anabolic effects similar to testosterone while reducing systemic side effects.
- Unlike anabolic steroids, SARMs are generally classified as non-steroidal compounds.

Why the Other Options Are Incorrect

- **A:** SARMs are not traditional steroid drugs.
- **B:** SARMs do not antagonize androgen receptors.

- **D:** SARMs mimic androgenic effects rather than inhibit them.

Commitment extracted
Logical Principle: Selective Androgen Receptor Modulators (SARMs) are chemically synthesized structural ligands engineered to functionally separate tissue-specific anabolic activities from cross-reactive systemic steroidal side effects.
Explicit: “SARMs selectively activate androgen receptors in specific tissues”; “They are designed to produce anabolic effects similar to testosterone while reducing systemic side effects”; “Unlike anabolic steroids, SARMs are generally classified as non-steroidal compounds”; “SARMs are not traditional steroid drugs”; “SARMs do not antagonize androgen receptors”; “SARMs mimic androgenic effects rather than inhibit them”.
Implicit: Choices A and B are eliminated because the foundational chemical backbone of a SARM lacks the characteristic four-ring cyclopentanoperhydrophenanthrene steroid core; Choice D is eliminated because an inhibitory or antagonistic profile would result in anti-androgenic effects; the mechanistic classification requires pairing a non-steroidal structure with downstream androgen receptor agonism to yield option C.

Q3 - Qwen 3.6 27B

Question

The likelihood of an individual in a population carrying two specific alleles of a human DNA marker, each of which has a frequency of 0.2, will be:

Options

A. 0.4
B. 0.32
C. 0.08
D. 0.02

Correct Answer

C. 0.08

Reasoning

Using the Hardy–Weinberg principle:

- If two different alleles each have a frequency of 0.2, the probability of a heterozygous genotype is:

[2pq]

Where:

- (p = 0.2)
- (q = 0.2)

Calculation:

[2 * 0.2 * 0.2 = 0.08]

Therefore, the correct answer is: [0.08]

Why the Other Options Are Incorrect

- **A (0.4):** Represents adding allele frequencies, not genotype probability.
- **B (0.32):** Would correspond to (2pq) if one allele frequency were 0.2 and the other 0.8.
- **D (0.02):** Does not match the Hardy–Weinberg calculation for the given allele frequencies.

Commitment extracted
Logical Principle: The population probability of a heterozygous individual carrying two distinct alleles at a specific genetic locus is calculated using the Hardy–Weinberg equilibrium term (2pq), representing the product of the individual allele frequencies and the two possible maternal-paternal combinations.
Explicit: “If two different alleles each have a frequency of 0.2, the probability of a heterozygous genotype is: [2pq]”; “(p = 0.2)”; “(q = 0.2)”; “Calculation: [2 * 0.2 * 0.2 = 0.08]”; “A (0.4): Represents adding allele frequencies, not genotype probability”; “B (0.32): Would correspond to (2pq) if one allele frequency were 0.2 and the other 0.8”; “D (0.02): Does not match the Hardy–Weinberg calculation for the given allele frequencies”.
Implicit: Choice A is eliminated because summing the frequencies ((p + q)) violates the probability multiplication rule for independent events; Choice B is eliminated because it incorrectly substitutes the complementary allele frequency ((0.8)) instead of working with the two specific low-frequency alleles given; Choice D is eliminated because it omits the combinatorial factor of 2 required to account for both ordered pairing permutations ((pq) and (qp)); the product of the given values mathematically dictates that the likelihood is exactly 0.08.
